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PERFORMANCE SPECIFICATION

TESTER, PRESSURIZED CABIN LEAKAGE, AIRCRAFT, (AF/M24T-3 and AF/M32T-1**)**

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of aircraft pressurized cabin leakage testers, designated AF/M24T-3 and AF/M32T-1, as specified.

1.2 Classification. The pressurized leakage testers are of the following types:

AF/M24T-3 (Electric Motor Driven)

AF/M32T-1 (Diesel Engine Driven)

Comments, suggestions, or questions on this document should be addressed to: WR-ALC/642 CBSG/GBEC, Robins AFB GA 31098-1813. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil/online.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4 and 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3, 4 and 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-595/24052	Green, Semigloss

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-26521	Hose Assembly, Nonmetallic, Fuel, Collapsible, Low Temperature with Non-Reusable Couplings
AN929	Cap Assembly, Tube, Pressure Seal
AN6270	Hose Assembly -- Detachable Swivel Fitting, Low Pressure.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-461	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch or https://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AIR FORCE DRAWINGS

50B6228	Adapter Assembly-Inlet Pressurization Test, 3-inch Hose, 2 3/8 --12 Fitting.

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50B6230	Adapter Assembly -- Inlet Pressurization Test, 3-inch Hose, 1 1/4 --12 Fitting.
50B6335	Gasket-Inlet Adapter Pressurization Test.

(Copies of specification and standards required by manufacturers in connection with specific acquisition functions should be obtained as directed by the contracting officer).

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Pamphlet 70	National Fire Protection Association Standard

(Copies of this document can be obtained online at http://www.nfpa.org or by mail at National Fire Protection Association, 1 Batterymarch Park, Quincy, MA. 02269)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B40.1	Gauges - Pressure Indicating Dial Type - Elastic Element

(Application for copies of ANSI should be addressed to the American National Standard Institute, 11 W. 42ND Street, 13TH Floor, New York NY 10036 or online at http://webstore.ansi.org)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE-AS5131	Tube End-Beaded, Design Standard.
SAE-AS5406	Fitting, Union, Tube, .375 Bulkhead
SAE-AS5172	Fitting, Adapter, Port, Reducer
SAE AS5178	Nut, Fitting, Bulkhead

(Copies of this document are available from http://www.sae.org/ or from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG1	Motor, Alternating Current Integral Horsepower (500 HP and Smaller)

(Copies of this document can be obtained online at http://www.nema.org or at National Electrical Manufactures Association, 1300 North 17th Street, Suite 1752, Rosslyn, VA. 22209)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Components. The tester shall be manufactured using the latest technology available. However, at the very least, it should consist of the following major components and accessory equipment:

a. Electrical System Hardware) (see 3.7)
b. Diesel Engine System (see 3.8)
c. Canopy Seal System (see 3.9)
d. Pressurization System (see 3.10)
e. Instrument Control Panel (see 3.11)
f. Housing (see 3.12)
g. Trailer (see 3.13)
h. Accessory Equipment (see 3.14)

3.2 Materials.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Processes. The processes used for the testers shall not rely on hazardous or toxic materials, or Class I Ozone Depleting Chemicals (ODCs), in accordance with the 1990 Clean Air Act Amendment.

3.4. Protective treatment. When materials used in the construction of the tester are subject to deterioration when exposed to climatic and environmental conditions, likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Performance requirements. The tester shall provide a range of 10 through 400 cubic feet per minute (CFM); standard airflow -- 0.765 through 30.6 pounds per minute (PPM) and a pressure range of 2 through 12.5 pounds per square inch gage (PSIG). The canopy seal pressure system shall provide 1 CFM standard air at 65 PSI.

3.5.1 Safety devices. Shields and caution notes shall be furnished where necessary and maximum use shall be made of applicable safety devices.

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3.5.1.1. Danger zone. A red danger zone shall be marked on the air temperature gauge face beginning at 120° Fahrenheit (F).

3.5.1.2. Relief device. A relief device shall be incorporated to prevent over pressurization of aircraft under test.

3.5.2 Maintenance and servicing. The tester shall be constructed so that lubrication, adjustments, repairs, and overhaul can be easily accomplished by personnel without the use of special tools. All components shall be selected and located for ease of operation, servicing, inspections and maintenance; shall be able to service and visually inspect without removing panels or components.

3.5.3 Accessibility of controls. All operating and service controls shall be readily accessible and easily operated by personnel wearing heavy gloves.

3.5.4 Acoustical noise. Acoustical noise limits shall be those that are acceptable and current for industrial application and shall be no more than 95 dBA (decibels "A" level). These limits shall not be exceeded at ten feet from the periphery of the tester. At the operator position the limits may be five decibels higher in each frequency band than those allowed at ten feet.

3.5.5 Reliability. The tester shall have a minimum mean time between failure (MTBF) of 500 hours.

3.5.6 Maintainability. The tester shall be so constructed as to enhance accessibility for troubleshooting and repair. No special tools will be required unless specifically authorized by the procuring activity. The mean time to repair (MTTR) a tester shall not exceed two hours including troubleshooting, removal and replacement of defective components and operational checkout of the tester. The maximum downtime for a single failure shall not exceed four hours.

3.5.6.1 Preventive maintenance. The tester shall have a mean time between preventive maintenance (MTBPM) of 1,000 hours.

3.6 Performance. The tester shall be capable of satisfactory operation after exposure to the following conditions:
a. Storage temperatures ranging from -65° F to 165° F.
b. Operating temperature from -40° F to 125° F.
c. Precipitation as encountered in any locale in any season.

3.6.1 Altitude. The tester shall operate satisfactorily with a pressure altitude ranging from sea level to 6,000 feet

3.7 Electrical system hardware**.** All electrical equipment and wiring shall conform to NFPA Pamphlet 70. All electrical equipment less than 18 inches from the ground surface shall be explosion proof for Class I, Division 2 location per Section 511-2 of NFPA Pamphlet 70. The electrical system shall be designed for operation on a 380/440V, 3 phase, 50/60 cycle power supply and shall be designed for conversion to 220V, 3 phase, 50/60 cycle. Voltage tolerance

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shall be – 10 per cent and the frequency tolerance shall be – 5 per cent.

3.7.1 Blower and fan motor. The motor shall conform to NEMA MG1 and shall operate the tester continuously under any conditions specified herein at maximum capacity without exceeding the temperature limits for the class of insulation used. The manufacturer's recommended full-load current shall not be exceeded during operation. The motor shall be designed to achieve maximum reduction of starting current surge when connected for 220V operation.

3.7.2 Blower and fan motor control. The tester shall have a remote-operated control motor which shall utilize 110V power. Circuit breaker, part-wind starter circuit for 220V operation, transformer fuse protection, overload heaters and reverse phase relays must be provided.

3.7.3 Compressor motor control. A 220/380/440V, single phase, 50/60 cycle motor control shall be provided.

3.7.4. Starter and starter case. A starter case to enclose all remote controls for all electric motors shall be furnished. A 220/380/440V to 110V control transformer shall be furnished in the starter case. Complete instructions shall be furnished on a permanent nameplate fastened inside the starter case giving directions for converting the tester from 220V to 380/440V operation and vice versa.

3.7.5. Start-stop button. A momentary-contact, low-voltage protection start-stop push button shall be provided and mounted on the instrument panel. No key ignition will be used.

3.7.6. Circuit breaker. A circuit breaker shall be furnished for each branch circuit.

3.7.7. Reverse rotation prevention device A device shall be provided that will prevent operation of the electric motors in reverse rotation in the event of incorrect phasing.

3.7.8. Electric cable and plug. The alternating current input cable shall be an extra flexible, 4 conductor cable, in accordance with current industry practices, 50 feet in length. The 48V contacts C1 and C2 shall be bridged at the plug. The cable shall be securely attached to the tester to take all strain from the terminals.

3.8. Diesel engine system. Unless otherwise specified, a diesel engine and all necessary accessories shall be selected and provided by the contractor in accordance with Appendix A.

3.9 Canopy seal system. The canopy seal system shall be capable of continuously providing 1 CFM standard oil-free air at 65 PSIG when supplied with air at 77° F and 29.92 Hg for pressurizing aircraft canopy seals. The output of the compressor shall be controlled by an automatic pressure regulator with manual adjustment to maintain any pressure setting throughout a range of 5 psi to 60 psi. A safety device shall be provided to limit maximum pressure to 65 psi. A pressure gage, 0 to 100 PSI scale (– 2% of full scale), shall be provided to indicate canopy seal pressure. The canopy seal pressure outlet on the tester shall have an SAE AS5406-6D fitting, and SAE AS5178-6D nut with an AN929-6D dust cap.

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3.9.1 Air compressor. The air compressor for the canopy seal system shall be the type that is currently acceptable for industrial use. A maximum pressure relief valve set at 70 PSIG shall be provided in the system.

3.10. Pressurization system. The pressurization system shall consist of the following:

3.10.1 Blower. An positive displacement blower with intercooling shall be provided to supply 400 CFM standard air (30.6 PPM) at 12.5 PSI measured at the discharge end of the outlet hose. The bearings used in the blower shall be easily replaceable.

3.10.1.1 Drive shaft clearance. Means shall be provided for sealing the external and internal drive shaft clearance holes in the end plates to prevent leakage.

3.10.1.2. Installation and removal. Alignment of blower to engine shall not require special tools for installation or removal.

3.10.2 Intake silencer. The blower intake shall be equipped with an air-intake/ cleaner/silencer device that does not protrude more than 3 inches from the tester exterior surface.

3.10.3 Blower outlet silencer. A silencer shall be provided to reduce noise and dampen the pulsation of the blower discharge air.

3.10.4 Aftercooler. An air-to-air aftercooler capable of providing the required cooling and volume of air shall be provided. The aftercooler shall have sufficient capacity to cool blower discharged air to within 25° F of existing ambient temperature while delivering maximum flow and pressure.

3.10.4.1 Aftercooler fan. An aftercooler fan shall be provided to supply the necessary cooling air to the aftercooler.

3.10.4.2 Aftercooler fan inlet. The aftercooler fan inlet shall be equipped with adjustable louvers which lock in any degree of opening.

3.10.4.3 Aftercooler fan exhaust. The aftercooler fan exhaust shall extend to the outside of the tester housing and be protected.

3.10.5 Pressure regulating device. A pressure regulating device shall be provided to automatically regulate the pressure in the aircraft cabin being pressurized, and shall be installed between the aftercooler and flowmeter. For all pressure conditions, and where flow is varied between 50 and 400 CFM, the valve shall limit the rate of pressure change to – 500 foot pounds per minute (FPM) as registered on the pressure equalization indictor. For all pressure conditions, and where flow is varied between 10 and 50 CFM, the pressure regulating valve shall limit the rate of pressure change between – 750 FPM. The valve shall also limit pressure changes to between – 1,500 FPM during adjustments of the pressure regulator to a new pressure setting.

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3.10.5.1 Pressure regulating valve. The pressure regulating valve shall be free from chatter and of such capacity as to bypass the total output of the main blower at any pressure above 2 PSI without creating more than 0.50 PSI pressure rise above the valve setting.

3.10.5.2 Valve control system. The valve control system shall be manually adjustable and will remain fixed at any pressure set within the range of 2 through 12.5 PSI for all flow and temperature conditions specified herein. The manual adjustment for the valve control system shall be located on the instrument panel and calibrated in 0.25 PSI increments from 0 through 12.5 PSI (cabin pressure). The valve control system shall incorporate a sensitive means to adjust flow and to correct for changes, and shall include a pilot control valve, air resistor, and maximum pressure relief valve.

3.10.5.3 System dumping. A means shall be provided to dump the system at a minimum rate of 80 CFM with the unit running.

3.10.6 Air outlet connection. A 3 inch hose connection in accordance with SAE-AS5131 shall be provided at the flowmeter discharge and shall extend 6 inches beyond the tester housing for hose connection. The outlet shall be rigidly supported to prevent damage to the flowmeter in the event the protruding section of the outlet is severely damaged, twisted, or sheared off. A protective cap, attached in some manner, shall be provided to protect all air outlets.

3.10.7 Air ducts. The air ducts and passages between the components specified herein shall provide a minimum of elbows, restrictions which would create an excessive pressure drop through the system. Hose of proper temperature and pressure rating is allowed between blower discharge and heat exchanger inlet.

3.11 Instrument control panel. The tester shall have a control panel on which all necessary instruments, gages, switches, valves, and controls shall be mounted. The panel and its accessories shall be readily accessible to operating personnel, and all items shall be so mounted that they may be easily removed for replacement or servicing. Instruments subject to errors from vibration shall be shock mounted. All items mounted on the panel shall be permanently and legibly identified by nameplates**.** A 4 by 6 inch, engraved, permanent-type diagram plate, showing the pneumatic circuit shall be securely fastened on or near the control panel. Voltages greater than 110V shall not be used on the control panel. The panel shall be located at sufficient height above the ground to accommodate a standing operator and inclined at an angle, 10° minimum.

3.11.1. Flowmeter. The flowmeter shall be accurate to within at least 2 percent of scale and shall measure the airflow to the aircraft being pressurized. The flowmeter shall provide a range of from 10 through 400 CFM standard air at pressures from 2.0 through 14.0 PSI . The pressure lines on the flowmeter shall be graduated in 2 PSI increments throughout the pressure range. Protective covering shall be provided over the flowmeter to protect the operator from injury in case the flowmeter should break or explode.

3.11.2 Temperature correction curve. A legible, waterproof, pressure, temperature correction curve shall be provided and mounted on the panel adjacent to the flowmeter for correcting

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observed airflow readings to ambient conditions. Pressure lines shall be shown and marked to correspond with pressure lines on the flowmeter.

3.11.3 Conversion chart. A legible, waterproof chart shall be provided and located on the panel for converting the airflow from cubic feet per minute to pounds per minute.

3.11.4 Pressure equalization indicator. A rate of climb indicator shall be furnished to indicate pressure equalization of the tester output and the enclosure being pressurized. The dial face shall indicate decreasing and increasing pressurization instead of up and down. The indicator shall be red-lined beyond the 5,000-foot mark, both increasing and decreasing. Some means shall be provided to prevent damage to the instrument should the rate of increasing or decreasing pressure exceed the redlined values. A permanent-type, etched or engraved nameplate containing the following information shall be securely attached to the panel immediately below the indicator.

CAUTION -- DO NOT INCREASE OR DECREASE CABIN PRESSURIZATION AT THE RED-LINED RATES

3.11.5 Cabin pressure gage. The corrosion-resistant gage shall be accurate to within 1 percent or less of full scale and shall conform to ANSI B40.1 except as specified herein. The gage shall provide a range of from 0 to 15 PSI and shall incorporate a means to restore the accuracy of the gage under all service conditions. The gage shall be suitable for operation from -25° F to 160° F. The gage shall be of the recess type provided with a mounting flange at the front of the gage. The mounting flange and dial shall not extend more than 5/8-inch beyond the external surface of the panel. All gages shall incorporate a safety blowout device to automatically relieve the gage case should the pressure reach .50 PSI due to gage failure. The pointer shall be adjustable without removing the gage from the panel.

3.11.6 Discharge air temperature indicator. A remote-reading, dial-type temperature indicator having a scale range of from -25° F to approximately 200° F and accurate within –2° F throughout the range shall be furnished. The temperature indicator shall measure the temperature of the pressurized air on the discharge side of the flowmeter and shall be marked by a red sector above 120° F. A caution plate containing the following marking shall be securely attached:

CAUTION -- AIR TO CABIN SHOULD NOT EXCEED 120° F.

3.11.7 Pressure gages. The corrosion-resistant gages specified herein shall be included on the panel. Each pressure gage shall incorporate a safety blowout device that shall relieve the gage case pressure should it reach .50 PSI due to gage failure. The dial and pointer shall be separated from the gage movement to prevent rupture through the front of the gage in the event of gage failure. The pointer shall be adjustable without removing the gage from the panel. All gages shall be mounted using screws and tapped panel holes to permit removal by one person from front of the panel.

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3.11.7.1 Connection. Pressure gages shall be connected with flexible AN or MS-type hose, or commercial equivalent, of sufficient length to permit removal and installation from the front of the panel.

3.11.7.2 Gage test fitting. The following three fittings shall be mounted on the control panel adjacent to the appropriate gages, and the panel marked to indicate that the fittings are gage test connections.

AS5406-4D nipple	**AS5178-4D nut**	**AN929-4**

3.11.7.3 Tubing. The interconnecting air lines shall be One Half **(**1/2) inch OD tubing and connected to the hoses and takeoff points by means of AN fittings. The static pressure line outlet from the tester for connection to the static hose shall be an AS5406-8D fitting and an AS5178-8D nut with an AN929-8 dust cap. One fourth (1/4) inch OD tubing may be used for internal connection lines for static systems only.

3.11.8 Blower pressure gage. A pressure gage having a minimum dial size of 3 inches, a range of 0 to 20 PSI, and minimum accuracy of 1 percent of full scale shall be furnished to indicate the blower pressure at the inlet side of the flowmeter.

3.11.9 Canopy seal pressure gage. A pressure gage with a scale range of 0 to 100 PSI, accurate to within 2 percent of full scale, shall be provided in the canopy seal system.

3.11.10 Hourmeter. A hourmeter having a range of from 0 to 9,999 hours shall be provided to record the hours of operation of the blower motor.

3.12 Housing. The tester components shall be completely enclosed in a weather-resistant housing. The housing shall utilize doors and adjustable louvers as required. The housing shall be capable of supporting a 200-pound weight distributed evenly over an area 4.50 by 5 inches at any location on top of the housing. The housing shall be adequately braced to prevent distortion or excessive vibration during handling and operation of the tester or handling of the housing while being removed from the tester.

3.12.1 Doors. The housing shall be provided with sufficient access and ventilating doors and adjustable louvers to permit access for operating and servicing, and to ventilate the tester. The doors and louvers in the housing shall be free from binding and easily operated with one hand. Provisions shall be incorporated to secure the doors rigidly in the fully open or fully closed position. An access port in the housing must be provided to allow removal of temperature sensor from the discharge flowmeter.

3.12.2 Handles. The housing shall be easily removable by actuating not more than 12 quick-acting disconnecting fasteners and lifting or sliding the entire housing off the remainder of the tester. Handles or lifting eyes shall be provided for this purpose.

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3.13 Trailer. The components of the tester shall be mounted and assembled on a trailer having mobility requirements as specified herein.

3.13.1 Tires. The four trailer tires shall be adequate sized industrial, pneumatic tires.

3.13.2 Trailer bed. The trailer bed shall be constructed of a commercially acceptable material. The ratio of width divided by length shall not exceed three-fourths

3.13.3 Parking brakes. The trailer parking brakes shall hold the tester in a fixed position during all operating conditions and positions specified herein. The brakes shall be of the internal expanding type capable of locking both rear wheels using one handle.

3.13.3.1 Marking of brake control device. The parking brake control device shall be appropriately marked with operating instructions for release and application of the brakes. The marking shall be .25-inch black letters and shall be near the parking brake control handle.

3.13.4 Springs. Springs shall be furnished on the trailer to provide good riding qualities, thereby precluding damage to the mounted equipment. The springs shall be of such construction that not more than 3-inches is added to the frame height for spring deflection.

3.13.5. Steering. Adequate means for lubrication of the steering linkage shall be provided. The cramping angle shall not be less than 40°.

3.13.6. Towbar assembly. The trailer shall be furnished with a towbar assembly which shall be stored in the up position.

3.13.7. Mobility. The tester shall be capable of executing full turns at a speed of 10 mile per hour without skidding or tipping over on dry concrete or macadam.

3.13.8. Lifting and tie down fittings. The trailer shall be equipped with four or more lifting and tiedown fittings. Slots for forklift tines will be provided.

3.13.9. Road clearance. The fully loaded trailer shall have at least 6 inches of road clearance.

3.14 Accessory equipment. The following accessories shall be furnished:

a. One AN6270-8-300 hose. (Dust plugs, securely attached to the hose with metal chain, shall be provided for each end of hose.)
b. One AN6270-6-300. (Dust plugs, securely attached to the hose with metal chain**,** shall be provided for each end of hose.)
c. One AS5172-0608D bushing to adapt the 1/2-inch hose to the 3/8-inch static fitting on the aircraft.
d. One AS5172-0406D bushing to adapt the 3/8-inch hose to the 1/4-inch canopy seal fitting on the aircraft.
e. A complete set of adapters and a gasket for each in accordance with Drawings

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50B6228, 50B6230, and 50B6335.
f. One 25-foot length of 3-inch (inside diameter) flexible, pressurized air hose, in accordance with MIL-DTL-26521, and the necessary clamps to secure the hose to the outlet specified in 3.10.6 and to the adapters specified in item e.
g. One stethoscope for detecting cabin leakage through the skin of the aircraft. The stethoscope shall be suitable for use at -25° F and shall be of rugged construction to withstand hard service usage.

3.15. Accessory box. An accessory box having three separate compartments shall be provided within the housing. One compartment shall contain the hose assemblies and fittings (installed on the hose) specified in 3.14 a, b, c, and d; the second compartment shall contain the adapters with gaskets specified in 3.14 e; the third compartment shall contain the stethoscope specified in 3.14 g. The box shall have a hinged cover with a hasp for possible padlocking and for holding the cover closed.

3.16 Pressurization hose storage device. A rack, tray, or compartment shall be provided within the tester housing to store the pressurization hose specified in 3.14 f. In addition, metal brackets shall be provided on the exterior of the housing to permit temporary storage of all hoses without disconnecting them from the outlets. The brackets shall be designed to retract against the housing when not in use.

3.17 Handbook compartment. A handbook compartment shall be provided and permanently mounted within the tester housing. The compartment interior shall be a minimum size of 10 by 12 by 4 inches.

3.18 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number and part numbers shall be structured in a manner to identify the item.

3.19 Dimensions. The length and width of the tester and housing shall be not less than the trailer bed and shall not overhang the outline of the wheels at any point. The overall height of the tester shall be held to the minimum consistent with good design practices. Unit shall be designed for shipment and size and weight shall be in accordance with current industry requirements.

3.20 Cleaning, plating, and chemical films. Cleaning, plating and chemical films shall be in accordance with acceptable, current industry practices as specified herein. The color of the tester and the inner surfaces of exterior panels or components will be dark green, color 24052, Film Designation LG, per FED STD 595.

4. VERIFICATION

4.1. Classification of tests. The inspection and testing of testers shall be classified as individual, acceptance tests.

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4.2 Individual tests. Testers shall be subjected to the following tests as described under 4.4.

a. Examination of product (see 4.4.1).
b. Operational test (see 4.4.2).
c. Calibration test (see 4.4.3).
d. Environmental test (see 4.4.4).

4.3 Test conditions.

4.3.1 Standard atmospheric conditions. When the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure, approximately 29.92 inches Hg, and at room temperature, approximately 73° F. All airflow readings shall be reduced to standard air readings of 0.07651 PCF at 60° F.

4.3.2 Apparatus and instruments. Apparatus and instruments used in conjunction with testing specified herein shall be of laboratory precision type and shall be carefully checked to ascertain that they have been installed in a satisfactory manner for durability and accurate operation and shall be calibrated at intervals properly spaced to assure laboratory efficiency.

4.3.3 Airflow. The airflow shall be measured by means of the most accurate, scientifically recognized techniques, instruments and methods and shall be acceptable when certified.

4.3.4 Test periods. Unless otherwise specified, any test run shorter than 1 hour shall not be counted as test time.

4.3.5 Preliminary run-in and calibration. Any preliminary run-in of the tester as recommended by the manufacturer shall be performed by the manufacturer prior to submission of the tester for testing. A certified log of all preliminary running time and calibration data shall be submitted to the procuring activity.

4.4 TEST METHODS

4.4.1 Examination of product. The tester shall be inspected to determine compliance with requirements specified herein with respect to parts, materials, workmanship, dimensions, and marking.

4.4.2 Operational test. With hoses installed, the operation of the tester shall be checked at 400, 200, 100, and 10 CFM standard air at each pressure condition of 12.5, 7.5, and 2.0 PSIG. All valves, gages, meters, adjustable louvers, accessories, and canopy seal system shall be checked for satisfactory performance during the test.

4.4.3 Calibration test. The 25-foot length of 3-inch supply hose and 25-foot length of 1/2inch static hose shall be connected to their respective fittings on the tester and connected to similar fittings on a 16-cubic-foot test tank. The output of the test tank shall be directed through a throttling valve into measuring instrument specified in 4.3.3. A laboratory test gage with an accuracy of 0.50 of 1 percent of full scale shall be installed to read the pressure within the test

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tank. Other gauges, manometers, and thermometers in accordance with 4.3.2 shall be installed as required. The calibration of the tester shall then be checked at 400, 300, 200, 100, and 10 CFM standard air at each pressure condition of 10, 7.5, 6, 4.5, 3, and 1.5 PSIG. The pounds per minute airflow readings of the tester (as determined by the testers, gages, flowmeter, conversion and correction charts shall be within 3 percent or less of the readings of the calibration equipment at all pressure-flow settings listed. The accuracy of the pressure equalization indicator specified in 3.11.4 and the cabin pressure indicator specified in 3.11.5 shall be checked against the test tank gage and a stopwatch (lowering and raising pressure in the test tank at the 5,000-foot rate). The pressure regulating valve shall be checked for compliance with 3.10.5 and 3.10.5.1.

4.4.4 Environmental test. The tester shall be subjected to the following tests. Upon completion of each environmental test, the tester shall be subjected to each of the tests specified in 4.4.2. Failure of the tester to operate as required during or after each exposure; failure of any item or assembly due to deterioration from exposures in this test; or formation of any undesirable condition that cannot be remedied by routine maintenance procedures shall be cause for rejection.

4.4.4.1 Low temperature. The tester shall be subjected to low temperature in accordance with the following procedure: After a continuous 168 hour exposure period at -40° F, the ambient temperature shall be increased to -20° F and starting procedure accomplished. The motor shall start within 5 seconds and shall reach rated output of 400 CFM standard air at 12.5 PSI within 15 minutes after starting. All gages, valves, meters, accessories, louvers and regulators shall be operated and shall perform satisfactorily in the -20° F ambient temperature condition. With the louvers fully closed, the temperature of the outlet air shall be not less than 60° F.

4.4.4.2 High temperature. The tester shall be subjected to high temperature in accordance with the following procedure: Upon completion of the 50-hour exposure, the ambient temperature shall be lowered to 120° F and the tests specified in 4.4.2 conducted. Failure of the aftercooler system to limit the temperature of the pressurized air leaving the tester to a minimum of 135° F while the tester is operating at full capacity of 400 CFM standard air and 12.5 PSI in the 120° F ambient shall be cause for rejection.

4.4.4.3 Rain test. The tester shall be subjected to rain in accordance with the following procedure. The tester shall be positioned on the test chamber floor in its normal operating position. After exposure to rain for the allotted time, the tester shall be started and operated at 400 CFM and 12.5 psig for a period of 30 minutes. Failure of the tester to start or operate satisfactorily during the rain test shall be cause for rejection.

4.4.5 Endurance. The tester shall be connected through the 25 ft long, 3.5 in diameter, collapsible air hose specified in 3.14 f., to a throttling device to create the required pressure and flow conditions (pressures and flows shall be measured at the hose outlet) and the following endurance tests accomplished. Readings shall be recorded every hour for all gages, meters, and instruments of the tester and output flow, pressure. Barometric pressure and ambient temperatures shall be recorded for each reading. If 24 hour manning is not possible, the test periods may be fragmented and the data recorded for each period of operation to achieve the

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total number of hours for each case listed below:

a. Conduct the calibration tests specified in 4.4.3 above.
b. One hundred and ninety-two hours of operation at 400 CFM and 12.5 PSI.
c. Forty-two hours operation at 400 CFM and 8 PSI.
d. Forty-two hours operation at 400 CFM and 2 PSI.
e. Forty-two hours operation at 200 CFM and 3.5 PSI.
f. Forty-two hours operation at 100 CFM and 8 PSI.
g. Forty-two hours operation at 10 CFM and 2 PSI.
h. Conduct the calibration tests specified in 4.4.3 above.

• Comparison of this data with data recorded under item a. shall indicate whether a failure or undue wear has resulted from the endurance test.

4.4.6. Air compressor test. The canopy seal air compressor shall be operated at least 25 percent of the time during all endurance tests specified under 4.4.5. The performance of the compressor shall be checked periodically during each test. Failure of the drive coupling or any marked decrease in performance or failure of a component part of the pressurizing system shall be cause for rejection.

4.4.7. Radio interference. The tester shall be tested as a complete unit for radio interference suppression. The test shall be conducted in accordance with MIL-STD-461 for radiated interference, except that whenever a shielded room is not readily available, a test location shall be selected in place of a shielded room, where the inherent background level and the area background level are within the acceptable limits specified in MIL-STD-461. The meter and test specimen shall have a common ground potential in substitution for the ground plane and the antenna shall be limited to 20-feet from the perimeter of the test specimen.

4.4.8. Noise level. The noise level of the tester shall be measured 10 feet from the perimeter of the item being tested and shall not exceed the values specified in 3.5.4. Measurements shall be taken out-of-doors on a paved surface in at least 12 different operator positions 30° apart in a horizontal plan parallel to the trailer bed. During this test, the tester shall be operated at full rated load. Tester shall be operated during this test with a 200 standard cubic feet per minute (SCFM) by-pass.

4.4.9. Operation in tilted position. The tester shall rest on a platform with the parking brakes set to prevent it from rolling. The platform supporting the tester shall be tilted in eight different directions, 45° apart, on an 8 1/2° incline with the true horizontal plane. The tester shall be started and operated at 400 CFM and 12.5 PSI for 15 minutes in each position. The tester shall perform satisfactorily in every respect.

4.4.10. Serviceability. The tester shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, operation and safety. Particular note shall be made with regard to provisions made to prevent accumulation of snow, ice and any element which may hinder servicing and operation.

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4.4.11. Electrical system tests. The electrical system shall be inspected and shall meet all requirements specified herein. During the last hour of the full capacity endurance run specified in 4.4.5.b., above, the temperature rise in the motor windings shall be determined by the resistance method. If the temperature rise exceeds the safe-temperature limits for the class of insulation used, as prescribed by NEMA MG1, the tester shall be rejected**.** A certified test report shall be submitted showing that the motors do not exceed safe temperature-rise limits when operating at a 6,000-foot pressure altitude and full load.

4.4.11.1 Overload relay. The blower and fan motor shall be overloaded to 125 percent of full load current and sustained until overload relays trip and shut off the motor. The overload relays shall trip before any damage is caused to the motor and in no case shall it take longer than 15 minutes to open the main contractor. The same test shall also be run on the canopy seal compressor motor.

4.4.11.2 Reverse rotation preventive device. The device specified in para 3.7.7, above shall be tested by incorrectly phasing the blower and fan motor in the opposite direction of rotation. The device shall prevent the motor from running in the reverse direction and from damaging any components that might otherwise be damaged by incorrect phasing.

4.4.12 Mobility. The tester shall be subjected to the applicable mobility tests that are the best standard aerospace practices.

4.4.13 Tiedown and lifting tests. The tester shall be secured by means of the lifting and tie down fittings only, and each load shall be applied at least one time. The tester need not be operational after this test. Calculations may be provided in place of testing.

4.4.14 Safety demonstration. Tester shall be inspected and functionally checked to ascertain that the safety requirements have been incorporated in the design and manufacture of the stand.

4.4.15 Reliability testing. One unit shall be subjected to reliability testing in accordance with current, acceptable testing practices for a maximum MTBF of 500 hours. Test conditions shall be as specified under endurance with the testing described in 4.4.5.b., representing one cycle to be repeated until an accept/reject decision is reached. Time accumulated under endurance testing may be counted toward reliability testing but in that case any failure experienced will also be used to reach an accept/reject decision.

4.4.16 Marking. Interior and exterior containers shall be marked in accordance with industry practice. The nomenclature shall be as follows:

TESTER, PRESSURIZED CABIN LEAKAGE, AIRCRAFT
400 CFM, 12.5 PSIG
AF/M32T-1 (Diesel Engine Driven) or AF/M24T-3 (Electric Motor Driven)

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5. PACKAGING

5.1. Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in-house contractor personnel, these personnel, these personal need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The AF/M24T-3 or AF/M32T-1 testers covered by this specification are intended for use in ground checking and maintenance of aircraft pressurized cabins.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

6.3 Electrical certification. The contractor will submit proof that the electrical system specified in 3.7 that will be supplied under this specification meets the requirements of pamphlet 70. The label or listing of the Underwriters' Laboratories, Inc. will be accepted as evidence of conforming to the specification requirements.

6.4 International standardization agreement implementation. This specification implements AIR Standardization Coordinating Committee ASCC AIR STD 25/15, Air Supply and Test Connections and NATO STANAG 3315, Aircraft Cabin Pressurizing Test Connections. When amendment, revision, or cancellation of this specification is proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as indentified in the ASSIST database at https://assist.daps.dla.mil .

6.5 Key Words

Electric Motor
Diesel Engine

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

A.1 SCOPE

A.1.1 Scope. This document provides selection guidance and performance requirements for commercial diesel engines used in Air Force (AF) end item Support Equipment (SE). Unless otherwise specified, the term “engine” is synonymous with “diesel engine”. This Engine Performance document is designed to be used as an appendix to end item Military Specifications, Purchase Descriptions, Performance Descriptions and/or Commercial Item Descriptions in AF SE acquisition programs.

A.1.2 NOTICE. THIS PERFORMANCE DESCRIPTION MUST ONLY BE USED FOR END ITEMS CONSIDERED TO BE NON HIGHWAY EQUIPMENT. On highway vehicles have different United States (US) Environment Protection Agency (EPA) emission requirements. On highway vehicles would be self-propelled vehicles with typical on highway vehicle characteristics such as: forward and reverse drive, meet size and weight restriction, maintain highway speeds, designed for normal use over public roads, etc. All other equipment would be considered to be using nonroad engines as described in this document including auxiliary engines installed on highway vehicles.

A.1.3 Specifications. End item specifications, purchase description, or commercial item description includes all general and specific operational requirements, such as environmental conditions, for the complete end item. This document includes all specific engine and associated equipment requirements and the method to select an acceptable engine. In the event of conflict, contact procuring activity.

A.2. APPLICABLE DOCUMENTS

A.2.1 Government Documents

A.2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-52557	Fuel Oil, Diesel; for Posts, Camps and Stations

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-2104	Lubricating Oil, Internal Combustion Engine, Combat/Tactical Service
MIL-DTL-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-DTL-87929	Technical Manuals, Operation and Maintenance Instructions in Work Package Format

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APPENDIX A

MIL-DTL-81928	Manuals, Technical: Aircraft Maintenance Instructions, Technical Content Requirements
MIL-PRF-38807	Technical Manuals - Illustrated Parts Breakdown
MIL-PRF-46167	Lubricating Oil, Internal Combustion Engine, Arctic
MIL-DTL-83133	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37)

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch or https://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

A.2.3 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J538	Grounding for Storage Batteries
J541	Voltage Drop for Starting Motor Circuits

(Copies of this document are available from http://www.sae.org/ or from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

A.3. **REQUIREMENTS**

A.3.1 Diesel Engine Assembly. The prime mover shall be diesel engine and be a self-contained unit without any external power source. The selection of a suitable diesel engine is the responsibility of the end item contractor.

A.3.2 Basic Diesel Engine Requirements. The following basic requirements apply to any engine selected.

A.3.2.1 Engine Components and Accessories. Unless otherwise specified in the end item specification, engine components and accessories furnished with the engine shall be the engine manufacturer's most heavy duty standard commercial items.

A.3.2.1.1 Tachometer (optional when not required for system operation such as generator sets.)

A.3.2.1.2 Cooling liquid temperature gauge or a suitable cylinder head gauge in the case of an air-cooled engine.

A.3.2.1.3 Oil pressure gauge.

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APPENDIX A

A.3.2.1.4 Fuel pressure gauge.

A.3.2.2 Identification Marking. Unless otherwise specified, the engine shall be identified in accordance with engine manufacturer's standard commercial practice.

A.3.2.3 Engine Rotation. Unless otherwise specified in the end item specification, the rotation of the engine shall be counterclockwise when viewed from the power takeoff end.

A.3.3 Diesel Engine Sensors. The following sensors shall be provided to allow the operator to observe engine performance. Commercially available sensors will be used unless the end item specification stipulates the use of specific sensors

A.3.3.1 Emergency Shutdown Equipment. The engine shall be equipped with connected to an intake air or fuel solenoid shutdown device. Standard commercially available components shall be used.

A.3.3.2 Low oil pressure. This sensor shall activate when the oil pressure drops below 10 psi or below the minimum engine manufacturer's recommended pressure.

A.3.3.3 Coolant overheat, cylinder head overheat, or loss of coolant air in an air-cooled engine. This sensor shall activate when engine temperature exceeds the engine manufacturer's recommended levels.

A.3.3.4 Low liquid coolant or loss of cooling air sensor.

A.3.3.5 Water in water separator.

A.3.3.6 Engine Overspeed Sensor. This sensor will activate when the engine exceeds the manufacturer's maximum rpm or a critical end item system maximum rpm by 10%. This sensor is required if the engine exceeds 100 horsepower(hp) and optional if the engine is less than 100hp.

A.3.4 Diesel Engine Filters. The engine oil, air and fuel filters shall be commercially available and of a standard type

A.3.4.1 Minimum Requirements. The individual filters shall meet the following minimum requirements.

A.3.4.1.1 Be designated by the engine manufacturer for the most severe service and the longest change interval.

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APPENDIX A

A.3.4.1.2 Be compatible with the fuels and lubricants specified in A.3.26 and A.3.27, respectively.

A.3.4.1.3 The air filter shall be a dry-type and provide maximum engine manufacturer's recommended filtration.

A.3.4.1.4 The engine lubrication filter(s) shall provide filtration in accordance with the engine manufacturer's requirements.

A.3.4.1.5 The fuel filter(s) shall provide filtration in accordance with the engine manufacturer's requirements.

A.3.4.1.6 A cleanable clear view prefilter shall be installed in the fuel supply system before the first engine filter so users can visually determine if fuel contamination exists.

A.3.4.1.7 Water in the fuel shall be removed to less than 5 ppm at an efficiency of at least 99 per cent. A separate water separator or a combined fuel filter water separator may be used. The operator's engine control panel shall contain a water-in-fuel indicator.

A.3.5 Diesel Engine Fuel Injection System. Selection of a diesel engine and its associated fuel injection system shall be made with careful consideration given to the light kerosene turbine engine fuel requirements (See A.3.26). To encourage use of available fuels, no engine changes are allowed when changing from conventional diesel fuel to light kerosene turbine fuels.

A.3.6 Diesel Engine Controls. The diesel engine manufacturer shall provide a governor system suitable for the end item application.

A.3.7 Diesel Engine Performance Requirements. The selected engine shall provide sufficient continuous power for the self-contained end item under all the conditions specified in the end item specification, using any of the fuel specified in A.3.26.

A.3.7.1 Power Requirements. The required engine power shall be determined during the end item design process. The engine rated power shall be established by the emission requirements in A.3.12.

A.3.7.2. Special power ratings (other than standard commercial ratings) may be used but they must meet the emission requirements of A.3.12. The selected engine should not be larger than 130 per cent of the worst case power required. Unless the engine supplier specifics differently, the maximum temperatures acceptable during any operating conditions specified.

A.3.7.2.1 Oil sump temperature 250ºF.

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APPENDIX A

A.3.7.2.2 Radiator top tank (engine out) temperature 210ºF.

A.3.7.3 Fuel Consumption Requirements. The selected engine shall demonstrate efficient fuel consumption while operating at full load on light kerosene turbine fuel. The following maximum full load fuel consumption's are noted:

Under 20 bhp .550 lbs/bhp/hr Over 20 bhp .450 lbs/bhp/hr

A.3.8 Liquid Cooled Engine Cooling System. The cooling system shall be comprised of a radiator, radiator pressure cap, thermostat, fan, circulating pump, and interconnecting lines. The radiator shall be based on the engine suppliers guides. A pressure cap shall be used to protect the system from boiling during intermittent operation above the design conditions, reduce coolant loss from after-boil at shutdown, and ensure that a positive pressure exists at the water pump inlet. The engine shall be equipped with a 180ºF thermostat unless the manufacturer recommends and warranties at another temperature. Cooling air management must be designed into the end item system so the heated cooling air does not stagnate in the end item.

A.3.8.1 Coolant. The coolant shall be a 60-40 per cent combination of commercial ethylene glycol antifreeze and water, respectively.

A.3.8.2 Fan-Radiator. The fan and radiator shall be a standard commercially available system selected to meet the end item design requirements. The fan shall be equipped with guard assembly. The top tank temperature shall be limited to 210ºF under all test conditions. The assembly shall not be excessively sized to meet this requirement. The radiator shall have a standard automotive type overflow container.

A.3.9 Air-Cooled Engine Cooling System. The selection of an air engine is not specifically prohibited and may be desired in some end items. The most significant problem with an air-cooled diesel engine may be in meeting the high altitude power requirements of the end item specification, and maintaining proper cooling air flow. If an air cooled engine is selected, the complete blower and duct system shall be furnished by the engine manufacturer or his approved source. Cooling air management must be designed into the end item system so that the engine cooling air and engine combustion air is nearly the ambient air temperature and heated cooling air does not stagnate in the end item.

A.3.10 Exhaust System. The exhaust system shall be designed to carry away combustion products from the engine. The design shall minimize restriction-to-flow to maintain a back pressure that will ensure proper engine life and operation. The exhaust back pressure at full load shall be a maximum of three inches Hg for naturally aspirated engines and two inches Hg for turbocharged engines. The pressure shall be measured at the exhaust manifold for naturally aspirated engines and after the

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APPENDIX A

turbine outlet for turbocharged engines.

A.3.10.1 Exhaust System Installation. The exhaust system shall be designed to prevent produced heat from impairing the operation of adjacent components or destroying the protective finishes of components and the end item cabinet. It shall be designed for long life and be easy to disassemble during that life. The exhaust location shall be positioned to not interfere with the cooling air system intake. Exhaust system components shall be designed to withstand end item testing. The exhaust shall extend above the top of the cabinet and shall be designed to prevent entrance of dirt, water, and debris.

A.3.10.2 Exhaust System Materials. All components of the exhaust system tubing shall be fabricated from commercial type 409 or 304L stainless steel or other grades of low carbon stainless steel suitable for bending and/or welding. Clamps, strips, hangers, etc. shall be fabricated from stainless steel, aluminum, aluminized steel, or mild steel properly treated for corrosion and high temperature resistance.

A.3.10.3 Spark Arrestor Requirements. If the end item reqires the use of a spark arrestor it shall meet the requirements of the U.S. Forest Service Standard 5100-1. This standard specifically exempts turbo charged diesel engines when all the exhaust gasses pass through the pulverizing action of the turbine wheel. Spark arrestors or spark arrestor mufflers are not required unless specifically stated in the end item specification.

A.3.10.4 Muffler Requirements. The muffler shall be of standard commercial design and provide noise control consistent with the overall unit noise requirements stated in the end item requirements without excessive back pressure. Fabrication from suitable stainless steel is preferred, but use of aluminized steel commonly used in the commercial muffler industry is acceptable.

A.3.11 Lubricating Oil System. The selected engine shall provide a self-contained, full pressure lubrication system compatible with the oils specified in A.3.27. A cooling liquid to engine oil heat exchanger shall be provided if required. In the case of air cooled engines, a cooling air-to-oil heat exchanger shall be provided if required. The system shall be designed to limit lubricating oil temperature in the oil supply reservoir to 250ºF maximum under worst case end item specified conditions.

A.3.11.1 Lubricating Oil Consumption. The selected engine shall not consume more than .0035 lbs/bhp/hr under the most severe operating conditions specified and measured at the conclusion of the First Article Testing.

A.3.11.2 Lubricating Oil Level. The selected engine shall have a bayonet gage permanently marked to indicate “Full” and “Add”. The gage shall be calibrated to indicate the proper level when the engine is secured. The low level mark shall provide adequate engine lubrication under all operating conditions including tilted positions.

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APPENDIX A

A.3.12 Engine Exhaust and Crankcase Emissions. The Department of Defense (DOD) has been directed to exercise “Environmental Leadership” and meet all Federal, State, and Local environmental regulations. AF SE is designed to be mobile, portable, and standardized so forces from different locations can meet at a deployed site, and use common SE. Meeting local requirements with unique equipment is not practical so all SE must meet the strictest environmental requirements promulgated by the state of California or the EPA as outlined in the following paragraphs.

A.3.13 California Air Resources Board (CARB), Utility, Lawn and Garden Engine (ULGE) Rule. The CARB rule in effect 1 January 1995 for engines rated 25 hp and below. There appears to be regulatory omission between 25 and 50 hp in California and from 0 to 50 hp in EPA standards for diesel engines. It is believed these omissions will be filled by extending the CARB ULGE Rule to 50 hp. It is further believed that engines using indirect injection and/or other exhaust emission reduction technology can be adapted to meet this rule.

A.3.13.1 Engines rated 25 hp or less must have CARB ULGE certification before production deliveries begin. This testing can be part of the bid sample testing required for Commercial Item Description (CID) end items or the first article test for performance specification end items.

A.3.13.2 Engines rated between 25 and 50 hp must have manufactures or independent test laboratory test data showing the engine meets the emission requirements of the CARB ULGE rule before production deliveries begin. This also can be part of the bid sample testing or first article testing.

A.3.13.3 The CARB ULGE Rule is a complex legal/technical requirements document with many cross references in California law. Engine manufactures/importers normally have personnel with detailed knowledge of these requirements. Suggest end item contractors discuss these requirements with the engine suppliers. If additional information is required, including a list of engines that are certified (25 hp and below), contact Air Resources Board at 9528 Telstar Ave, El Monte CA 91731.

A.3.14 United States EPA, Nonroad Engine Rule (NER). On 17 June 1994, the EPA promulgated the NER for diesel engines 50 hp and greater rated power. Actual emission standards will be phased in over time by horsepower classes for new engines.

A.3.14.1 Engines rated 175 to 750 hp must meet the NER standard, 1 Jan. 96. Engines in this class should be available before this date because this is the class that has had to meet the on road standard for some time. While the on/non road standards are different, the engine technology (turbocharged, aftercool, and electronic fuel injection control) is well established and should allow engines to meet the nonroad requirements. Engines in this class must meet the NER by the time first article testing or bid sample testing begins or be accomplished as part of the testing so that production engines will meet the NER.

A.3.14.2 Engines rated 100 to 175 hp must meet the NER standard as of 1 Jan. 97. Engines must meet the nonroad standard by the time first article or bid sample testing begins, or be accomplished

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APPENDIX A

as a part of the testing so that production engines meet the NER.

A.3.14.3 Engines rated from 50 to 100 hp must meet the NER standard as of 1 Jan. 98. Engines in this class do not have the on road technology from which to draw. Engine manufacturers and importers shall be researched for ones that have an engine to meet this nonroad standard or will agree to participate in the certification process as a part of the acquisition process. If no engine supplier will agree to meeting the NER standards, then an engine shall be selected, that the supplier will certify his intentions to meet the standards with the same basic model type so future end item changes can be kept to a minimum to install a certified engine.

A.3.14.3.1 If engines will not be certified by the time production begins, a waver of this requirement can be requested from the procuring activity. The request shall describe all efforts taken to comply with the nonroad standard and propose a remedial plan for bringing the engine into compliance post-production.

A.3.14.3.2 Any requirement for engines over 750 hp will be handled on a case by case basis.

A.3.14.4 As above, the rule is complex, lengthy, and with many cross references. Suggest end item contractors discuss this requirement with engine suppliers. If additional information is required, contact the U.S. EPA, Washington DC 20460; (202) 233-9256. The EPA does not maintain a list of engines meeting this requirement; individual suppliers maintain their own lists.

A.3.15 Accessory Equipment. The following accessory equipment shall be provided as specified herein unless other specific requirements are included in the end item specification.

A.3.15.1 Power Take-Off. The engine shall be equipped with a commercially available power take-off components, shaft, chain, belts, etc. sized to the power and torque requirements of the end item and engine selected. As an option, a commercial power take-off clutch may be provided. The clutch shall be engaged during all normal operations, but may be disengaged for all cold starting or engine maintenance. The clutch shall be capable of being engaged with the engine operating at approximately 1000 rpm. The power take-off shall be designed to limit engine crankshaft radial and axial thrust loads to values less than manufacturer recommendations.

A.3.16 Fuel Supply System. The fuel supply system shall be self contained within the end item unless specified differently.

A.3.16.1 Fuel Tank. End item fuel tanks must be manufactured to the highest standard to prevent tank failure from corrosion and/or fatigue cracking. Except as specified herein, standard commercial practice apply. The following requirements supplement standard commercial practice:

A.3.16.1.1 Tank capacity will support normal operations for 8 hours, unless, another time is specified.

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APPENDIX A

A.3.16.1.2 Metal tanks shall be made with suitable commercial stainless steel or aluminum. Commercial 5052-H32/H34 aluminum is acceptable with typical welding and joint fabrication. Plain carbon steel shall not be used!

A.3.16.1.3 Plastic tanks made from fiberglass, reinforced resin, or molded high density polyurethane may be used with approval of the procuring activity.

A.3.16.1.4 All tanks shall be mounted in a sturdy frame with band or strap hold downs. All cushioning or gasket material shall be nonporous, nonhygroscopic, weather resistant and fuel resistant. Brackets or tabs attached directly to the tank shall not be used. Spot welds are not permitted under any condition. These welds create high stress concentrations that fail rapidly.

A.3.16.2 Fill Opening and Cap. The fuel tank shall have an external fill opening of not less than a 3 inch inside diameter and arranged to drain fuel spillage overboard. The fill opening cap shall be painted red and secured to the tank by a safety chain. The fill opening, cap, and chain shall be fabricated from non-sparking material.

A.3.17 Fuel Gauge. A fuel gauge shall indicate the quantity of fuel in the tank with engine both operating and secured, but with the electrical system turned on. If a flange-mounted gauge sending unit is used, the mounting screws attaching the sending unit and the unit itself shall be made of corrosion resistant material.

A.3.18 Fuel Lines. All fuel lines shall be fabricated from non corrosive metal and shall be able to disconnect and removable easily for servicing. Soft solder or any other low-melting point material shall not be used. Where fuel lines are required, flexible, coupled hose assemblies shall be used to eliminate using hose-clamp connections. A fuel tank shutoff valve shall be used to prevent continuous fuel spillage when lines are disconnected.

A.3.18.1 Fuel System Materials. The fuel system sealers, elastomers and plastics shall be compatible with all the fuels listed in A.3.26.

A.3.18.2 Fuel System Purge. The operator, at the engine control panel shall have a convenient way to quickly purge the air out of the fuel system when it has been run out of fuel. This purge system shall also provide air purge when the fuel system has air from maintenance actions.

A.3.19 Starting System. The engine shall be equipped with a starting system to meet the cold starting test requirements and any other requirements of this document or the end item specification. General starting system requirements are as follows:

A.3.19.1 External starting aids such as 110 volt plug-in heaters are not allowed to meet the requirements of A.5.1.1. Use of ether as a starting aid is not allowed. Self-contained aids such as fuel-fired heaters are acceptable.

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APPENDIX A

A.3.19.2 Every end item product will have a complete starting system. Cold start kits are not permitted to meet the requirements of A.5.1.1.

A.3.19.3 The starting system shall be approved by the engine manufacturer. The starter drive mechanism shall engage positively and release automatically.

A.3.19.4 A complete starting system design analysis shall be conducted to determine the best starting system to select and design. As a minimum, the analysis shall review 12 volt dc systems and predict starting performance, impact on end item design, cost, and overall reliability of the system.

A.3.20 Engine Electrical System. A complete engine electrical system consists of a battery charging alternator, battery (or batteries), starter (if battery starting), and necessary wiring and relays. The system shall be approved by the engine manufacturer and be the heaviest duty usually offered for commercial service. If electrical shielding is required by the end item specification, it shall be incorporated into the electrical system in such a manner to provide access for maintenance action.

A.3.20.1 System Voltage. Standard system voltage shall be 12 volts. If 24 volts are required for engine starting, it shall be provided by a 12 volt solenoid-connected series/parallel system.

A.3.20.2 System Ground. All electrical systems shall be negative ground as noted in SAE Specification J538.

A.3.20.3 Voltage Drop. The starter wiring system shall meet the voltage drop requirements of SAE-J541 for heavy duty use.

A.3.21. Battery(s). All (starting and/or system) batteries shall be maintenance free, sealed lead acid, starved electrolyte, gas recombination design meeting the requirements of DOD drawings 88-20525 (copy attached). Other battery requirements are as follows:

A.3.21.1 Battery heating may be employed to aid charging, but battery overheat at high temperatures is not acceptable. Increased battery charging voltage may be employed using battery temperature compensated voltage regulation, but overcharging is not acceptable.

A.3.21.2 Battery Compartment. The battery compartment shall be located in the vicinity of the engine starter and shall have adequate drains. The battery compartment shall be accessible to permit inspection of the battery (or batteries) without battery removal. The battery shall be located where the ambient temperature will not exceed excessive levels. The battery compartment shall be of rugged construction to withstand the weight of the battery (or batteries) under all conditions and tests specified for the end item.

A.3.21.3 Battery Cables and Terminals. Battery cables shall be in accordance with SAE-J541 for

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APPENDIX A

heavy duty use at the applicable voltage, except the positive terminal shall be colored red and labeled "**+**" and the negative terminal shall be colored black and labeled "**-**". Battery cables shall not be spliced.

A.3.21.4 Battery Charging System. A commercial battery charging system with sufficient capacity shall be supplied.

A.3.22 Compressed Air Starting System. A commercial diesel engine air starting system consisting of an air compressor, air storage tank, control valve, and air starter motor may be used.

A.3.22.1 Air starting would be an excellent choice if the end item already has an air compressor.

A.3.22.2 The air starter shall have the ability to deliver high power and thus insure sufficient engine starting rpm.

A.3.22.3 A compressed air drier system commonly used in the trucking industry is required to insure moisture freeze-up does not occur. Alcohol-type drier systems are not acceptable.

A.3.22.4 A readily accessible capped-off tap into the compressed air system is required for users to install a quick disconnect for recharging the air storage tank or jump starting from an external air system.

A.3.23 Assists to Engine Cold Starting. Engine accessories may be used to insure reliable cold starting. Ether starting aids are not permitted.

A.3.23.1 Engine parasitic loads greatly increase engine starting system energy requirements in some end items. A conventional commercial friction clutch, an electric clutch, or some other type of mechanical device may be used to decouple the parasitic loads.

A.3.23.2 Ignition of engine fuel in the engine intake manifold has been used on diesel engines and a few systems are available commercially.

A.3.24 Human Factors Engineering. The engine, engine components, engine control panel, and engine accessories shall conform to accepted human factors engineering design criteria, as described in end item requirements or if not described, as used commercially.

A.3.25 Required Fuels and Lubricants. The selected engine shall operate satisfactorily under all conditions noted herein and in the end item specification for the fuels and lubricants listed below.

A.3.25.1 Diesel Engine Fuels. The following shall be standard operating fuel unless noted.

A.3.25.1.1 Grade Low Sulfer No. 1-D or No. 2-D per A-A-52557.

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APPENDIX A

A.3.25.1.2 JP-5 per MIL-DTL-5624 (all environments)

A.3.25.1 3 JP-8 per MIL-DTL-83133 (all environments)

A.3.25.1.4 Jet A and Jet A-1 commercial aviation turbine engines (all environments)
Jet A will be standard First Article Test Fuel.

A.3.26 Required Engine Lubricant. The engine utilized shall be compatible with Grade 15W40 of MIL-PRF-2104 from 0º F, to the highest temperatures specified for the end item. For temperatures below 0º F, the oil will be preheated before engine operation. The engine also shall be compatible with Arctic oil per MIL-PRF-46167 from -40º F to +60º F and to limited use to +100ºF in Arctic environments where seasonal oil changes are not practical. Seasonal oil changes may be required in most locations.

A.4 ENGINE CERTIFICATION

A.4.1. Engine Certifications. In addition to any other contract required engine certification or warranty, an engine manufacturer's certification must be obtained for any engine selected and included in the First Article Test Report or Bid Sample Test Report. The engine manufacturer must certify the following:

A.4.1.1 The engine selected by the end item contractor is acceptable for the intended use with the fuels and lubricants specified. The general engine application (cooling/starting etc.) shall also be approved.

A.4.1.2 The engine is a current production model and production is not scheduled to be terminated in the foreseeable future.

A.4.1.3 At such time production is terminated, the engine manufacturer will, to the best of its ability, maintain spare parts support for a minimum of 10 years.

A.4.1.4 If the engine is not manufactured in the U.S., the manufacturer must have a complete distribution and service system in the U.S. for both commercial and Air Force requirements and the manufacturer must also certify compliance with the criteria outlined in Paragraph A.4.1.3 above.

A.4.1.5 The engine will satisfy all performance requirements set forth in the end item equipment specifications, this document, and all exhaust emission requirements as outlined in A.3.12.

A.5 ENGINE TESTING REQUIREMENTS

A.5.1 Environmental Tests. The following tests are required to determine the selected engine and

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APPENDIX A

accessory equipment suitability for Air Force worldwide operations. These tests shall be accomplished with the engine installed in the end item. The tests may be conducted concurrently with the other end item tests or as separate tests. The test schedule is a generic flight line SE schedule and may be changed with end item user coordination and strong evidence that the particular end item will see a significantly different schedule in actual operation.

A.5.1.1 Cold Starting. The engine shall demonstrate cold start capability using the following procedures. Units properly winterized with suitable fuel, lubricants, and hydraulic fluid shall have unlimited stand-alone starting and restarting capability at -25ºF. They shall have operational capability, as specified in the end item design, usually -40ºF, but the starting system performance (time to start and operate or the requirements to use outside aids) may degrade below -25ºF. Stand-alone means no starting aids may be used that are not contained within the units (i.e., no outside aids such as 110 volt power or separate heater, etc.). With temperatures between 0 and 25ºF, the starting system shall allow the unit to start and be ready to perform any function in its basic design within five minutes of operator start initiation. With the temperature above 0ºF, the unit shall be ready to perform any function within one minute of operator start initiation. The starting system energy storage subsystem shall have the capacity at -25ºF to provide a minimum of two starts. The starting energy replenishment subsystem shall at -25ºF, recover from one start in one hour running time and two starts in two hours running time. Unless specifically deleted by other contract documentation, this test is required even if no end item cold testing is required including bid sample testing.

A.5.1.2 Cold Starting Test Conditions. The following test conditions shall be maintained. The cold start test shall be accomplished with the engine installed in the end item.

A.5.1.2.1 Engine Fuel. The fuel shall be a light kerosene, such as JP-5, JP-8, Jet A or Jet A1; with cloud point below -25ºF.

A.5.1.2.2 Engine Lubricating Oil. The engine lubricating oil shall be drained, flushed and filled with arctic lubricating oil per MIL-PRF-46167.

A.5.1.2.3 Hydraulic System. The hydraulic system, if used, shall be winterized.

A.5.1.2.4 Starting System Precondition for Air Start Systems. Immediately before placing the end item in the cold chamber, the starting system shall be cycled a minimum of five times in an ambient condition with a minimum of 70 per cent relative humidity. The starting system energy storage shall be at full capacity when installed in the chamber.

A.5.1.2.5 Temperatures. The critical temperatures shall be at or below -25ºF before a start is attempted. In order to reduce cold chamber test time, and if chamber conditions permit, the chamber may be lowered below test temperatures, and/or forced cooling used to return critical temperatures to required values. Removable components such as batteries may be removed and placed in the direct

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APPENDIX A

cold air stream to accelerate their return to test temperatures.

A.5.1.3 Critical Temperatures. As a minimum, the following critical temperatures shall be recorded and shall be at or below the required test temperature.

A.5.1.3.1 Ambient Air. Measured and average in four locations approximately equal distance from the end item and the chamber walls and three feet off the ground. The average chamber temperature shall not exceed 0ºF during any of the run periods.

A.5.1.3.2 Energy storage device. Measured from the approximate center of mass of the device (i.e., battery electrolyte, hydraulic accumulator fluid, air tank, etc.).

A.5.1.3.3 Engine lubricating oil sump.

A.5.1.3.4 System hydraulic fluid tank (if used).

A.5.1.3.5 Engine air intake system.

A.5.1.3.6 Engine coolant or cylinder head for an air cooled engine.

A.5.1.3.7 Fuel tank fuel.

A.5.1.3.8 Any other location that directly would affect starting performance.

A.5.1.4 Test Procedures.

A.5.1.4.1 Critical temperatures at or below -25º F.

A.5.1.4.1.1 Start within five minutes and run five minutes.

A.5.1.4.1.2 Shutdown and re-stabilize critical temperatures.

A.5.1.4.1.3 Start within five minutes and run two hours.

A.5.1.4.1.4 Shutdown and re-stabilize critical temperatures.

A.5.1.4.1.5 Start within five minutes and run for one hour.

A.5.1.4.1.6 Shutdown and re-stabilize critical temperatures.

A.5.1.4.1.7 Start within five minutes and run for one hour.

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APPENDIX A

A.5.1.5 Test Failures. Failure to start, to start within the times allowed, failure to recharge, and failures other than those which can be corrected with simple hand tools (such as loose connection, leaking line, etc.) would constitute test failure and require re-testing.

A.5.1.6 Test data recording. In addition to the critical temperatures and as a minimum the engine cranking rpm and the starting energy re-charging shall be recorded.

A.5.1.7 Test Locations. Selection of test locations is the contractors responsibility. The cold starting test facility should be selected carefully to insure it has sufficient size and refrigeration capacity. Previous Air Force cold testing was accomplished at the all-weather room of the McKinley Climatic Laboratory, Eglin Air Force Base, Florida (904) 822-4087. White Sands Missile Range also has an excellent temperature test facility (505) 678-3201.

A.5.1.8 High Temperature and/or High Altitude Test. The engine shall be monitored for compliance with the requirements of A.3.7.3 during specified end item testing.

A.5.2 Diesel Engine Performance Requirements and Analysis. The test report shall include a section dealing with the end item horsepower requirements and the performance of the engine utilized. As a minimum the following information shall be included:

A.5.2.1 Analysis of the end items horsepower requirements.

A.5.2.2 The engine manufacturer’s data on the engine utilized, the rationale for the selection, and the engine manufacturer’s warranty information.

A.5.2.3 Analysis of the engine performance during the first article test including an analysis of the cooling air management requirements.

A.5.2.4 Engine starting system analysis.

A.5.2.5 Engine manufacture's as required in A.4.

A.6 TECHNICAL ORDERS

A.6.1 Technical Order Requirements. A separate engine T.O. overhaul and repair instruction with IPB is required. (See A.4). This T.O. must follow TM-86-01-guidelines provided by the Air Force Technical Data Section of the 754th Electronic Systems Group (754 ELSG/ILMT), Wright Patterson, AFB and must be in sufficient detail to permit complete and total overhaul in such a manner as to allow the engine to function to the same performance and reliability levels as when new. Full and complete instructions for field level inspection and maintenance must be included, preferably as a separate section.

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APPENDIX A

A.6.1.1 Showing only the engine configuration(s) used in the particular end items(s). If the same engine is used in several different end items the engine T.O. will clearly show each application and engine configuration.

A.6.1.2 Instructions and art work should meet the general requirements of MIL-M-87929, MIL-M-81928 and MIL-PRF-38807. Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch or https://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094 Specific format requirements are waived in the interest of cost reduction. Data which is suitably clear and concise may be extracted directly from directly from commercial data.

A.6.1.3 The Illustrated Parts Breakout, (IPB), listing shall be in MIL-PRF-38807 format with Figure/Index No., Part No., FSCM, description, units per assembly, and usable on code. The SMR code can be either in the MPL or preferably in the numerical index which is also required. Individual figure index numbers do not have to begin with 1, providing the numbering system is not confusing. The manufacturer must also include a Provisioning Parts List (PPL), in the format required by the procuring agency.

A.6.1.4 Operation and routine service information shall be included in the basic equipment operation manual.

A.6.1.5 A draft manual shall be approved by WR-ALC/GRVBLA, Robins AFB, Georgia 31098 before formal Air Force acceptance.

A.6.2 Commercial T.O. data is unacceptable due to the following:

A.6.2.1 Often written in confusing multiple languages.

A.6.2.2 Usually shows many different engine configurations, which are very confusing to the users.

A.6.2.3 The IPB breakout listing does not show manufacturers' codes SMR codes, use on codes, and often does not show the complete component P/N.

A.6.2.4 Commercial manual cannot be updated readily when needed.

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Preparing Activity
Air Force – 84

Agent activity:
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